

RegionOS — Product Description (Alpha Development)

RegionOS is a visual monitoring and automation platform that transforms your desktop into a collection of intelligent screen regions. Instead of capturing an entire display, users can create customizable "regions" that track specific areas of their screen. Each region acts as an independent window into their digital environment, capable of monitoring, analyzing, and interacting with the content inside it.

Users can create 2, 4, 8, 16, or unlimited regions depending on their needs. Each region can be assigned a purpose: monitoring a health bar in a game, tracking a stock price, reading chat messages, watching a terminal output, analyzing a dashboard, or feeding information into an AI assistant.

RegionOS gives users control over **what their computer sees** by allowing them to define exactly which parts of the screen matter.

Core Concept

"Your screen is made of information. RegionOS lets you isolate, monitor, and understand it."

A Region is a user-defined rectangle placed anywhere on the screen.

Example:

```
+-----+
|      Monitor      |
|                   |
| +-----+ +-----+ |
| | Region 1 | | Region 2 | |
| | Discord | | Stock Chart | |
| +-----+ +-----+ |
|                   |
| +-----+ +-----+ |
| | Region 3 | | Region 4 | |
| | Game HUD | | Terminal | |
```

| +-----+ +-----+ |
+-----+

Each Region can independently:

- Capture its assigned area
 - Display a live preview
 - Analyze visual changes
 - Extract text
 - Detect objects
 - Trigger actions
-

MVP (Version 0.1 Alpha)

1. Region Creation System

Users can:

- Click "Create Region"
- Drag a box over any area of their screen
- Name the region
- Save the position and size

Example:

Region Name:

"Enemy Health Bar"

Position:

X: 520

Y: 840

Size:

300 x 40

2. Region Dashboard

Main application window:

RegionOS

[+ New Region]

Region 1

[Live Preview]

Status: Monitoring

FPS: 30

Region 2

[Live Preview]

Status: Monitoring

FPS: 30

Users can see all active regions in one place.

3. Screen Capture Engine

Requirements:

- Capture only selected areas
- Adjustable refresh rate
 - 1 FPS
 - 5 FPS
 - 30 FPS
 - 60 FPS

Each region runs independently.

Example:

Region 1 → 60 FPS (Game HUD)

Region 2 → 1 FPS (Stock price)

Region 3 → 10 FPS (Chat)

4. Region Tools

Each region can have settings:

Basic

- Rename
- Resize
- Move
- Hide
- Pause

Monitoring

- Detect change
- Detect color
- Detect image
- Detect text

AI

- Describe region
- Summarize changes
- Extract information

Phase 2 Features

OCR

Example:

Region:

\$4,532.20

RegionOS:

Detected:

Price = \$4,532.20

Change:

+\$45.20

Change Detection

Example:

A game boss health bar:

Before:



After:



RegionOS:

Change detected:

Health decreased 30%

Automation

Rules:

WHEN:

Region "Login Button"

changes to blue

THEN:

Click location

or:

WHEN:

Stock price > \$500

THEN:

Send notification

Advanced Vision Features

AI Vision Mode

A user selects a region:

"Analyze this"

AI response:

This region contains:

- A Discord message
- Username: Alex
- Timestamp: 3:42 PM
- Message indicates a meeting request

Region Groups

Example:

Gaming setup:

FPS Game

Combat Group:

- Health
- Ammo
- Map
- Enemy

Communication Group:

- Discord
- Team Chat

Target Users

Gamers

- AI game assistant
- HUD tracking
- Strategy tools

Developers

- Monitor logs
- Debug applications
- Watch builds

Traders

- Multiple market feeds
- Price alerts

Streamers

- Chat monitoring
- Stream dashboards

AI Users

- Give AI controlled vision without sharing entire screens

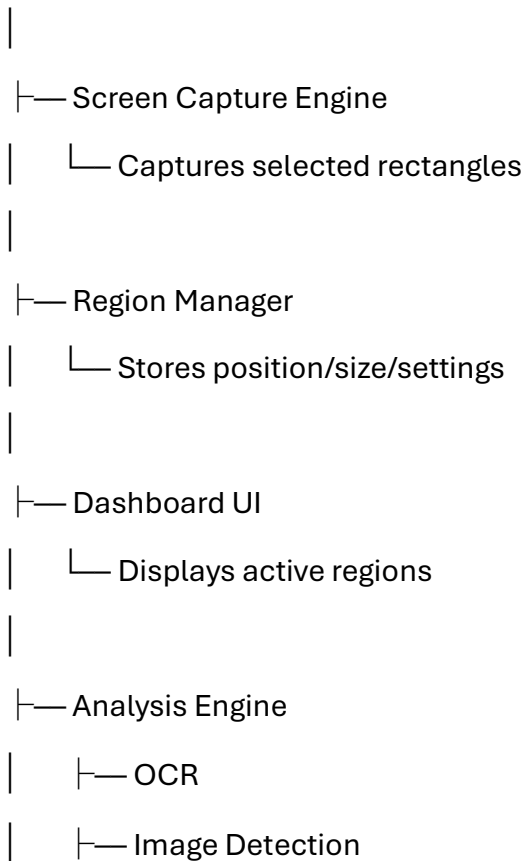
Businesses

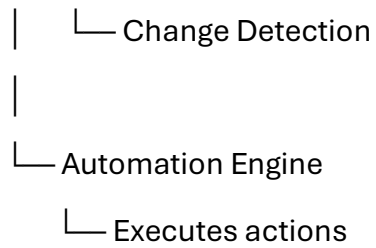
- Legacy software automation

Technical Starting Point

Alpha Architecture

RegionOS





Alpha Goal

The first working version should simply achieve:

"I can draw multiple boxes on my screen, and RegionOS continuously displays and updates those sections independently."

Everything else builds from that.